

## Literature Review Structure

- The literature review was mapped directly to the architecture of our proposed system.
- Papers were categorized into five technical layers: GAN-based generative design, LLM persona simulation, AI agent interaction, feedback optimization, and application systems.
- To ensure balanced understanding, each team member reviewed **11 papers covering all domains**.
- The findings were then consolidated and presented according to the **system architecture pipeline**.

One liner summary : Existing literature extensively explores generative design using GANs and human behavior simulation using LLMs. However, very few systems integrate these capabilities into a unified feedback-driven simulation framework capable of evaluating product concepts before launch.

Paper Distribution:

| Member    | Paper Numbers | GAN (3)    | LLM Persona (3) | AI Agents (3) | Application (1) | Feedback / Optimization (1) |
|-----------|---------------|------------|-----------------|---------------|-----------------|-----------------------------|
| Sejil     | 1–11          | 1, 2, 3    | 4, 5, 6         | 7, 8, 9       | 10              | 11                          |
| Samanyu   | 12–22         | 12, 13, 14 | 15, 16, 17      | 18, 19, 20    | 21              | 22                          |
| Sarvagnya | 23–33         | 23, 24, 25 | 26, 27, 28      | 29, 30, 31    | 32              | 33                          |
| Shlok     | 34–44         | 34, 35, 36 | 37, 38, 39      | 40, 41, 42    | 43              | 44                          |

Domain Coverage Summary:

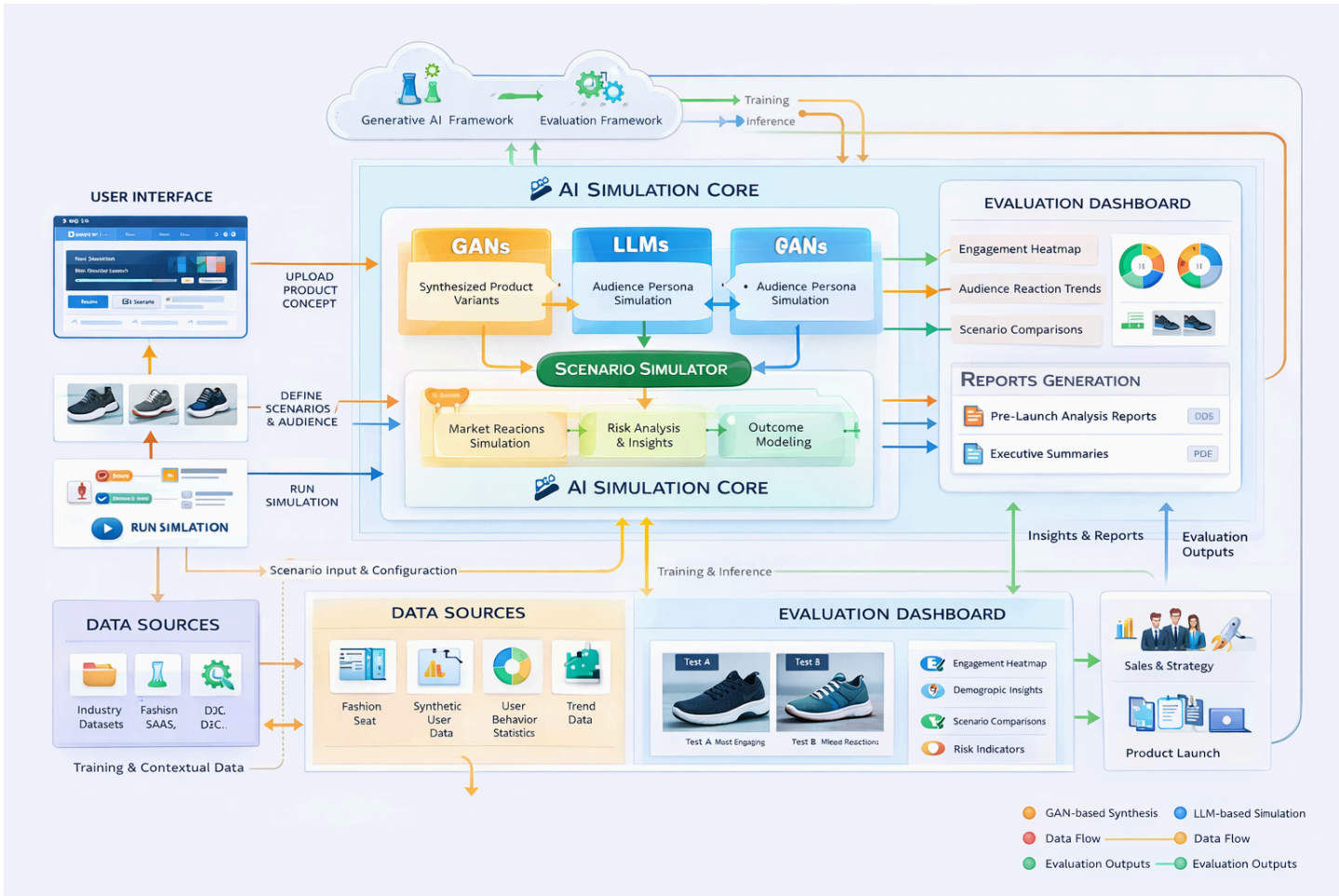
| Domain                  | Total Papers |
|-------------------------|--------------|
| GAN / Generative Design | 12           |
| LLM Persona Simulation  | 12           |
| AI Agents               | 12           |
| Application Systems     | 4            |
| Feedback Optimization   | 4            |
| Total                   | 44           |

Research Gaps Identified:

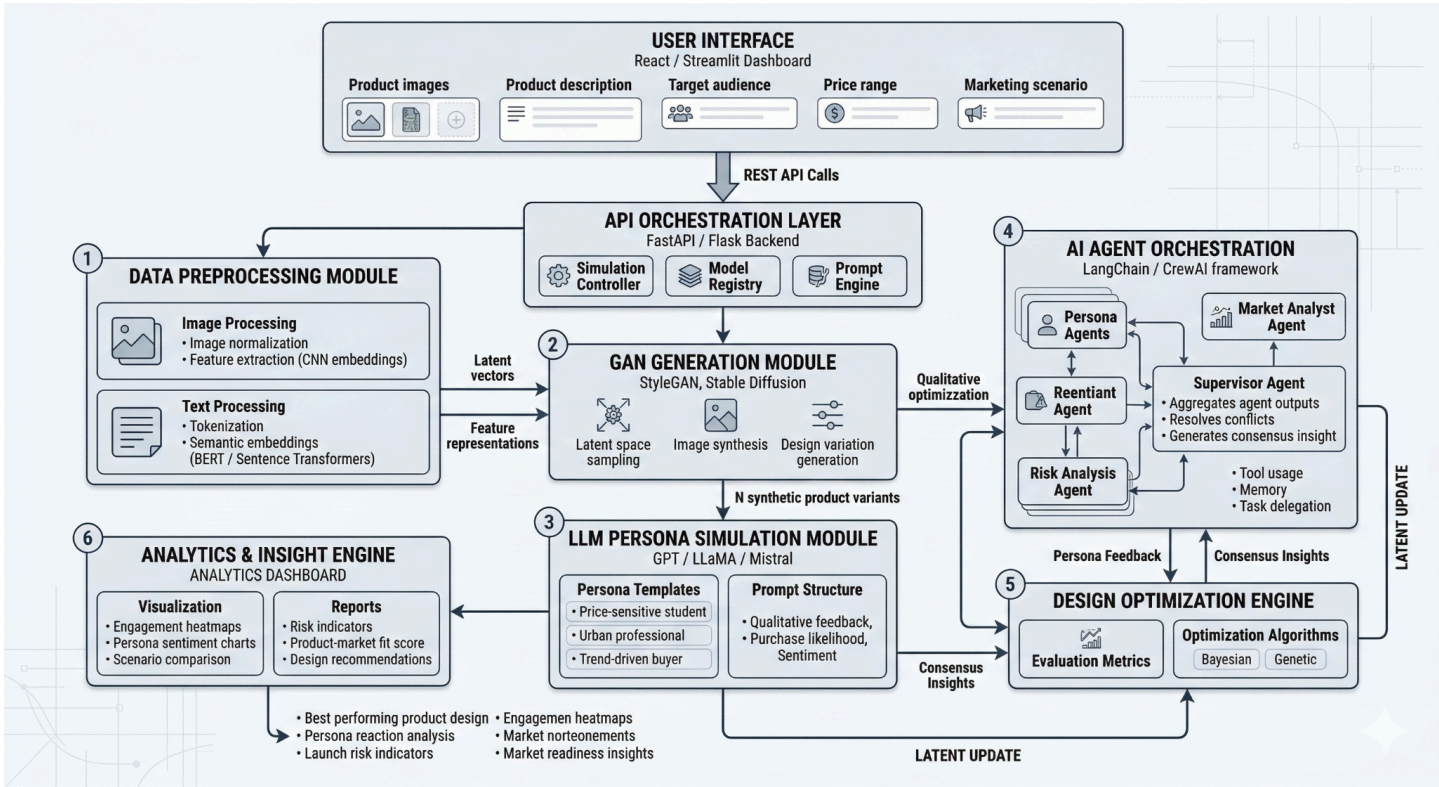
| Member    | Domain                      | Research Gaps |
|-----------|-----------------------------|---------------|
| Sejil     | AI agents                   |               |
| Samanyu   | LLM Persona                 |               |
| Sarvagnya | GAN                         |               |
| Shlok     | Feedback loop + Application |               |

System Architecture: (Closed-loop generative optimization architecture)

Phase 1:

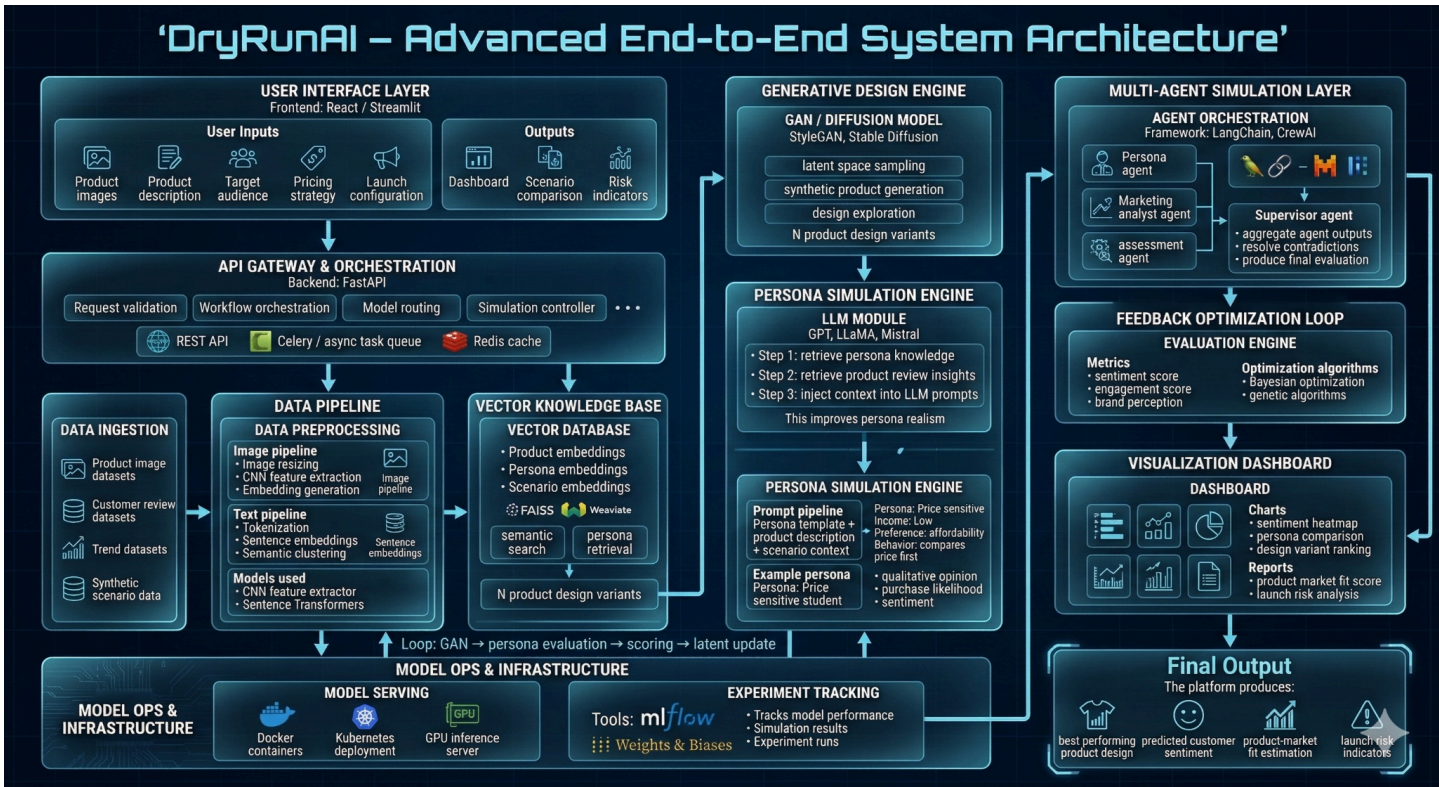


Phase 2:



The system integrates generative AI, persona simulation, and agent-based reasoning into a feedback-driven optimization pipeline. Product variants are generated using GANs, evaluated by simulated audience personas through LLM agents, and iteratively optimized using scoring signals to approximate product–market fit before launch.

Phase 3:



# DryRunAI: Simulated Product Testing and Optimization

